THE WHITE ARMY

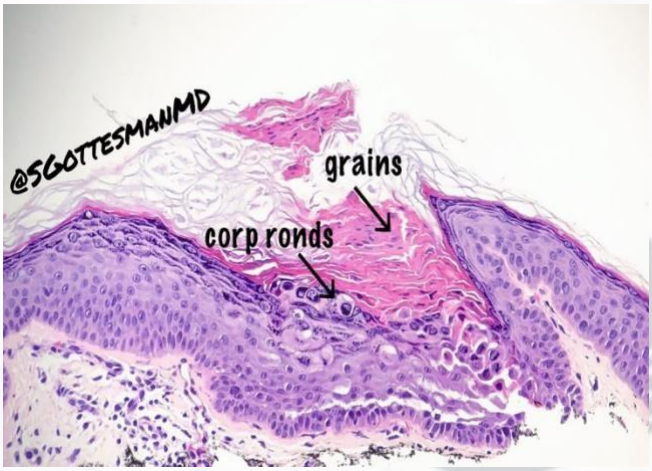
EPIGNOSIS

NATIONAL LEVEL ONLINE DERMATOLOGY QUIZ

PRELIMS -QUESTIONS and ANSWERS.

DATE-16/08/2020

1. A young male presents with multiple dirty, warty papules over the seborrheic areas of the body,biopsy from the lesion shows the following findings.What is your diagnosis?



A) Hailey hailey disease

B) Dariers disease

C) Lichen planus

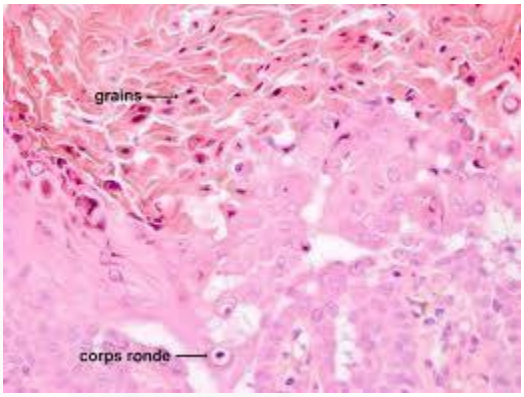
D) Psoriasis

✅ Option B –Darrier's disease

History of characteristic multiple dirty warty papules over the seborrheic areas is suggestive of Darrier's disease.

- The diagnosis is confirmed by the presence of suprabasal acantholysis and dyskeratosis. Dyskeratosis (faulty and premature keratinization) results in ***corps ronds and grains.***
- ***Corps ronds*** are round cells with central pyknotic nucleus that is surrounded by clear halo and peripheral dyskeratotic material. Present in spinous and granular layer.
- ***Grains*** are flattened cells with elongated nucleus present in stratum corneum.

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2.A 34 year old man who returned from a trip, presented with a history of fever and malaise. He recalled being bitten by an insect which was identified by the locals as tsetse fly. On examination, characteristic lymphadenopathy along the posterior cervical lymph node chain was found. What is the most likely diagnosis?



A) Filariasis
B) African sleeping sickness
C) Borreliosis
D) Loa loa disease

✅Option B- African sleeping sickness

- WINTERBOTTOM'S SIGN: enlargement of lymph nodes in the posterior cervical chain.
- It is seen in early stages of African trypanosomiasis.
- In this question, patient gives history of travel which suggests an endemic infectious disease.
- History of insect bite suggests the role of a vector. Tsetse fly is known vector for trypanosomiasis.

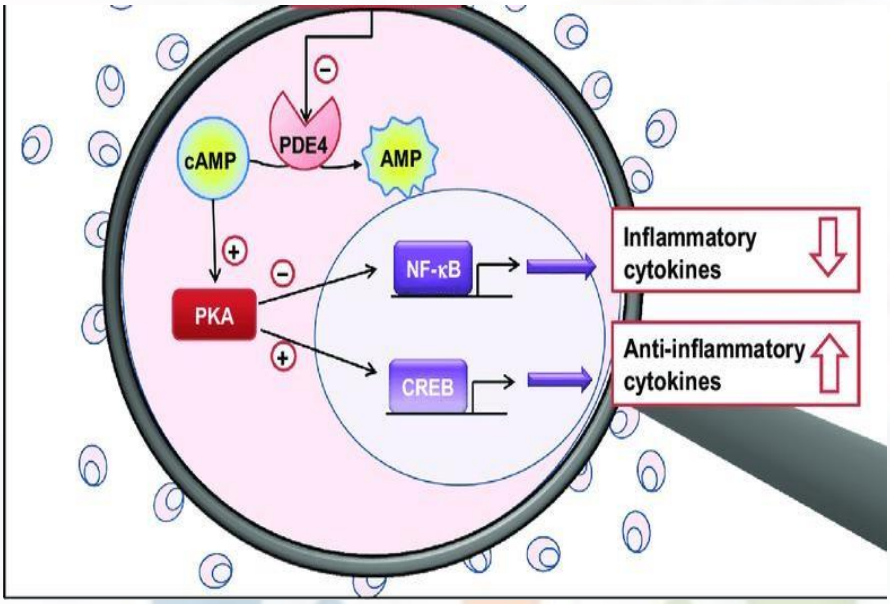
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- Filariasis also has fever, lymphadenitis and lymphangitis leading to swollen and edematous arms or legs. But the characteristic posterior cervical chain involvement is not seen. It is also not transmitted by tsetse fly.
- Loa loa disease: vector is Chrysops and has characteristic calabar swellings.
- Borreliosis- lyme disease is a tick borne disease. Skin findings include erythema migrans (producing targetoid lesions) and acrodermatitis chronica artrophicans.

3. Which drug used for psoriasis inhibits Phosphodiesterase (PDE) 4?

A) Methotrexate

B) Clobetasol



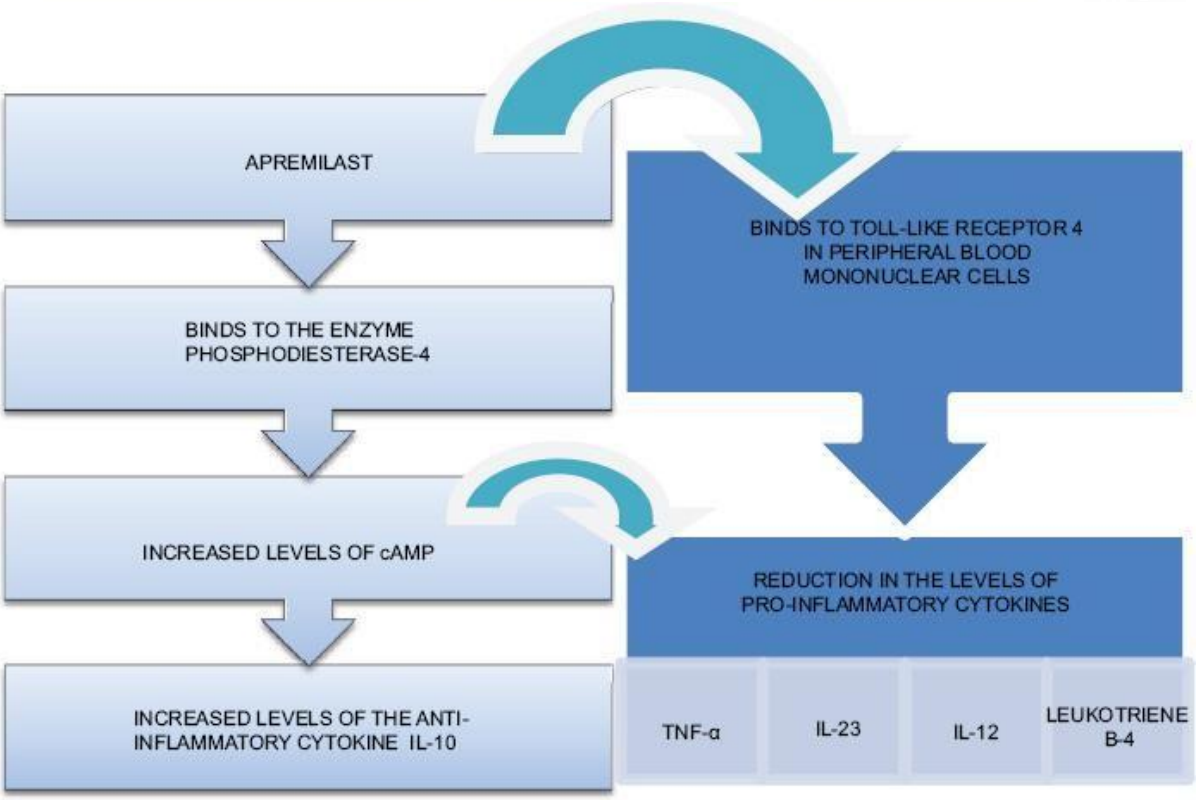
C) Cyclosporine

D) Apremilast

✅ Option D- Apremilast

- Apremilast is an oral PDE-4 inhibitor used for psoriasis.
- PDE-4 is concentrated in inflammatory cells, natural killer cells and keratinocytes.

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4) This characteristic sign is seen in?



A) Lichen planus
B) Pityriasis rubra pilaris
C) Psoriasis
D) Onychomycosis

✅ Option C- Psoriasis

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- The characteristic finding seen in this question is SALMON PATCH/OIL DROP.
- It is the orange-red discolouration of the nail.
- It is a manifestation of nail bed involvement in psoriasis.

Anatomical site affected	Clinical outcome
Proximal germinative matrix	Pitting, Beau's lines, onycholysis, leukonychia
Distal germinative matrix	Focal onycholysis, leukonychia, red lunulae, nail plate thinning
Whole matrix	Crumbling of the nail plate
Nail bed	"Oil drop sign," onycholysis, splinter hemorrhages, subungual hyperkeratosis, nail discoloration
Nail folds	Loss of cuticle, paronychia

5.This condition arises from?



A) Ectopic apocrine glands
B) Ectopic eccrine glands
C) Ectopic sebaceous glands
D) Ectopic apo-eccrine glands

✅Option C – Ectopic sebaceous glands

- The condition in the image is Fordyce's spots.
- They are seen commonly found on the lips, buccal and genital mucosa.

Box 37.1. Modified Sebaceous Glands

- Meibomian glands of eyelids
- Tyson's glands of prepuce
- Free sebaceous glands on the mucocutaneous surfaces of female genitalia
- Montgomery's tubercles on areola of nipples
- Ectopic glands in cervix uteri, tongue, and parotid glands
- Margins of lip (Fordyce's spots)

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6. This type of Alopecia seen in

A) Alopecia areata

B) Trichotillomania

C) Androgenic alopecia

D) Lichen plano pilaris

✅ Option D – Lichen plano pilaris

- The image shows a scarring type of alopecia which can be identified clinically by the affected region showing loss of follicular openings and the skin being shiny and smooth.
- All the other options are causes of non-scarring alopecias.

7. The auto-antibody that is classically associated with drug induced lupus erythematosus is

A) Anti-centromere antibody

B) Anti dsDNA antibody

C) Anti histone antibody

D) Anti Sm antibody

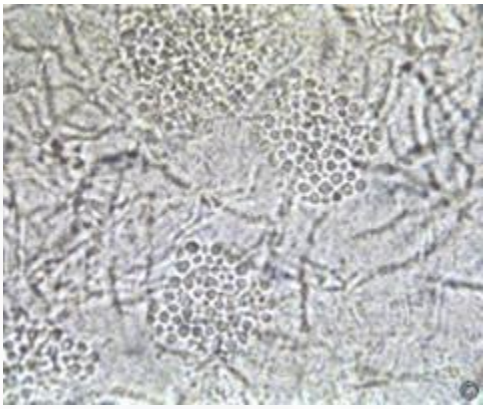
✅ Option C- Anti histone antibody.

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Antibody	Disease Association
Anti-IgG (Rheumatoid Factor) Anti-CCP (Cyclic citrullinated peptides)	Rheumatoid Arthritis
Anti-Nuclear antibodies (ANA) Anti-dsDNA Anti-Sm (Smith antigen)	Systemic Lupus Erythematosum (SLE)
Anti-Histone	Drug induced Lupus
Anti-SSA (Ro) & SSB (La)	Sjogren syndrome, SLE
Anti-U1RNP (ribonuclear protein)	Mixed connective tissue disease, SLE
Anti-Phospholipid	Antiphospholipid syndrome, SLE
Anti-Smooth Muscle	Chronic Autoimmune Hepatitis
Anti Centromere	Limited Scleroderma (CREST)
Anti- Scl-70 (DNA Topoisomerase I)	Diffuse Scleroderma
Anti-Mitochondrial	Primary Biliary Cirrhosis
Anti-Thyroid Peroxidase (Microsomal) Anti-Thyroglobulin	Hashimoto Thyroiditis
Anti-Jo-1 (Histidyl tRNA synthetase)	Dermatomyositis & Polymyositis
Anti-Basement Membrane	Goodpasture Syndrome
Anti-Gliadin IgA Anti-Endomysial IgA Anti-Transglutainase IgA	Celiac Sprue
Anti-Myeloperoxidase (P-ANCA)	Microscopic polyangiitis Churg-Strauss Syndrome
Anti-Proteinase 3 (C-ANCA)	Wegener Granulomatosis
Anti-Parietal cell & Intrinsic factor	Autoimmune gastritis
Anti-Glutamic acid decarboxylase	Diabetes type 1
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8) A young male presented with multiple, scaly hypopigmented patches over back. Woods lamp showed yellow fluorescence. On examination of scales with KOH the following findings was seen. What is your diagnosis?

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A) Tinea versicolor
B) Tinea corporis
C) Dermatitis artefacta
D) Erythrasma

✅ Option A – Tinea versicolor

- The question describes a classical clinical picture of T. versicolor with yellow fluorescence on Wood's lamp.
- On KOH examination the classical ***Spaghetti and meatball appearance (Banana and grape appearance)*** is seen. The slender septate hyphae represent the spaghetti/banana and the spores represent the meatball/grapes.

9) 1. Anagen is the growing phase of hair

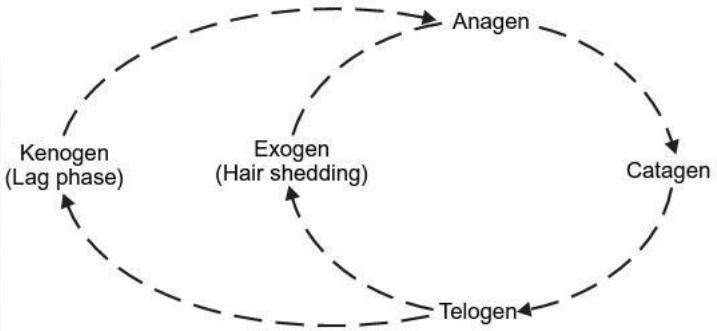
2. Catagen is the resting stage of hair
3. Stem cells are present in hair bulb
4. Lanugo hair is present in fetus

A) Only 1 is correct
B) 1 and 2 are correct
C) 1 and 4 are correct
D) 1,2,3,4 are correct

✅ Option C

- Anagen is the growing phase of hair
- Catagen is the involuting phase.
- Telogen is the resting phase.
- Stem cells are present in the **BULGE** area and NOT bulb area.
- Lanugo hair cover fetal skin and are shed at 8-9 months or just after birth.

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Sl no.	Stage	Duration	Percent of hair
1	Anagen phase	2 and 7 years	80%–90%
2	Catagen	2–3 weeks	1%
3	Telogen	3 months	8%–9%

10) A 1 year old male child presented with itchy scaly plaques and papules all over the body, predominantly over the seborrheic areas. Electron microscopy showed tennis racket shaped structures within the cells. The IHC marker most likely to be positive is:

A) CD20
B) CD14
C) CD123
D) CD1a

✅ Option D –CD1a

- The question describes a child with Langerhan histiocytosis (Letterer Siwe type)
- ***Birbeck granules*** are small "tennis racket" shaped membranje bound granules typically seen in Langerhans histiocytosis.
- IHC markers for these cells are CD1a, CD 207/Langerin.

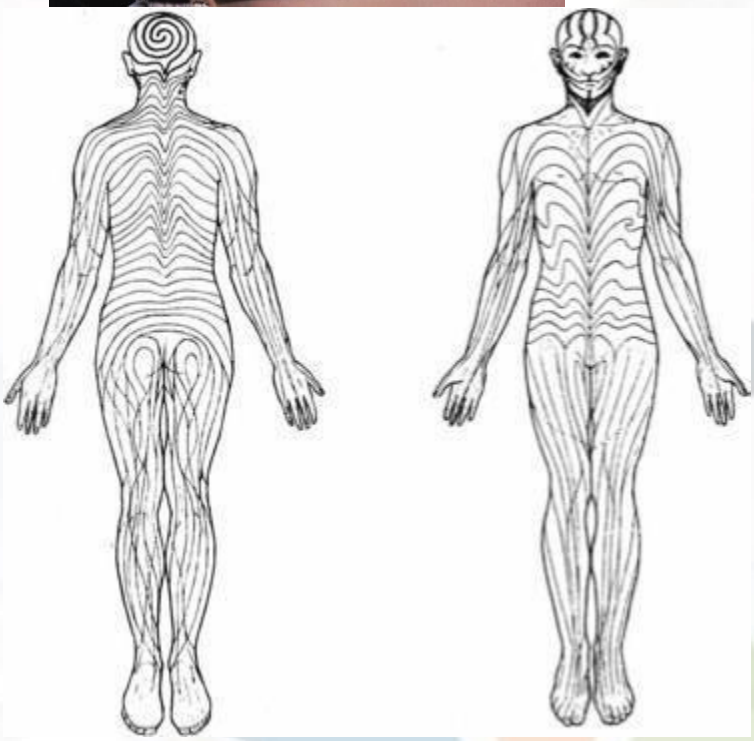
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11) A 7 year old child presented with linear band over the right arm. The lesions follows which of the following lines?



A) Resting skin tension lines
B) Cleavage lines
C) Pigmentary demarcation lines
D) Blaschkos lines
✓

Option D – Blaschko's lines



- The condition shown in the image is lichen striatus.
- Blaschko lines represent the developemental growth pattern of skin.
- Do not correspond to any nervous/vascular or lymphatic structures.
- Resting/relaxed skin tension lines are lines that lie perpendicular to long axis of underlying facial muscle. Incisions should be placed parallel to these lines for minimal scrarring.
- Pigmentary demarcation lines/ Futcher's lines/ Voigt's lines are physiological abrupt transitions from hyperpigmentation to hypo or normal pigmentation. 8 types have been described.

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12) A 21-year-old previously healthy man presented with weight loss, sinus congestion, ear pain, cough and hematuria. Oral examination of a patient revealed the following image what is your likely diagnosis?



A) Granulomatosis polyangiitis
B) Scarlet fever
C) Kawasaki disease
D) Eosinophilic polyangiitis.

✓ Option A –Granulomatosis polyangiitis 8

- The question describes the classical triad of GPA/Wegner's granulomatosis which is upper respiratory tract involvement (sinus congestion and ear pain), lower respiratory tract involvement (cough) and renal involvement (hematuria).
- Classically ***STRAWBERRY GINGIVA*** is associated with this condition.
- Eosinophilic polyangiitis aka Churg Strauss syndrome is characterized by asthma, eosinophilia, vasculitis and neuropathy.
- STRAWBERRY TONGUE is seen in scarlet fever and Kawasaki disease.



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13) child presents with fever and vesicles all over body. The casusative agent is likely?

A) HHV1
B) HHV2
C) HHV3
D) HHV4

✓ Option C –HHV 3

- The classical appearance shown in the question is "dew drops on a rose petal"
- It is seen in varicella infection in the vesicular stage caused by human herpes virus 3.

Human Herpes Virus

HHV-1	Herpes Simplex Virus 1	(HSV-1)	α
HHV-2	Herpes Simplex Virus 2	(HSV-2)	α
HHV-3	Varicella Zoster Virus	(VZV)	α
HHV-4	Epstein-Barr Virus	(EBV)	γ
HHV-5	Cytomegalo Virus	(CMV)	β
HHV-6	Human Herpes Virus 6	(HHV-6)	β
HHV-7	Human Herpes Virus 7	(HHV-7)	β
HHV-8	Kaposi's Sarcoma-associated Herpes Virus (KSV)		γ

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14) A pregnant women is diagnosed with anogenital warts. All of the following can be used for her treatment except?

A) Tricholoroacetic acid
B) Podophyllin
C) Cryotherapy
D) Radiofrequency

✅ Option B – Podophyllin

- It is an anti-mitotic agent and even has anti-viral properties.
- Acts by destabilizing the microtubules during cell division and causes metaphase mitotic arrest.
- It is contraindicated in pregnancy because it can cause limb malformations, septal heart defects, polyneuritis and pre auricular skin tags in the fetus.
- All the other modalities can be used in pregnancy.

15) A 42 year old male who is a known case of lepromatous leprosy presented with history of sudden appearance of crops of evanescent pink coloured tender papules and nodules over the legs, face and trunk. It was associated with fever and joint pain. Your most likely diagnosis would be :

A) Erythema nodosum leprosum
B) Reversal reaction
C) Erythema induratum
D) Lucio phenomenon

✅ Option A – Erythema nodosum leprosum

- The question classically describes the lesions of ENL/type 2 lepra reaction. It is a Type 3 hypersensitivity reaction.
- Reversal reaction is type 1 lepra reaction which is usually seen in BORDERLINE group of patients (BT, BB, BL). Presents with inflammation of pre-existing patches that become red and swollen and neuritis.
- Lucio phenomenon is a reactional state that occurs in lucio leprosy. Presents with redish blue plaques that ulcerate with irregular margins. Usually patient is afebrile and nerves are not affected.
- Erythema induratum is a type of panniculitis that also presents with nodules that ulcerate but is a reaction due to M. tuberculosis.

16) (i) Headlight sign is seen in atopic dermatitis
(ii) Atopic march consists of atopic eczema, allergic rhinitis and asthma
(iii) Dupilumab , used for atopic dermatitis targets IL-6
(iv) Hanifin and Rajka criteria is used for Atopic dermatitis

A) All the above statements are true
B) Only i & iii are true
C) Only i, ii & iv are true
D) Only i & iv are true

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✅
Option C

- Dupilumab is a new biologic that is approved for atopic dermatitis(AD) that **targets IL-4 & IL-13** which play a key role in the pathogenesis of AD.
- Other new drug is **CRISABOROLE 2%** ointment which is a PDE-4 inhibitor.
- All the other statements are true.

17) An 8 year old child presents with tense vesicles arising at the periphery of old lesions, located in the extremities and genital area, with no other significant diseases. What is the most likely diagnosis?



A)Bullous pemphigoid
B)Pemphigus vulgaris
C) Linear IgA bullous dermatoses
D)Dermatitis herpetiformis

✅ Option C- Linear IgA bullous dermatosis

- The history suggests tense vesicles and bullae which indicates a sub-epidermal blistering disorder. The image of the classical ***"string of pearls/cluster of jewels" appearance*** indicates Linear IgA dermatosis.

18) An elderly female presents with a depigmeted patch over the forehead .What is the most likely allergen associated with this condition?



A) para-tertiary butyl phenol
B) para-tertiary methyl phenol
C) ortho-tertiary butyl phenol
D) ortho-tertiary methyl phenol

✅ Option A- Para-tertiary-butyl phenol (PTBP)

- The adhesive material in bindhi contains PTBP which causes contact leucoderma.

19) Urinary porphyrin levels are elevated in all of the following except?

A) Porphyria cuatnea tarda
B) Congenital erythropoetic porphyria
C) Erythropoetic protoporphyria
D) Acute intermittent porphyria

✅ Option C – Erythropoetic protoporphyria

- As the porphyrinogens progress through the heme synthesis pathway, the number of carboxyl side groups decreases ❼ this brings down the water solubility ❼ decreases excretion through urine.
- Erythropoetic protoporphyria occurs due to ferrochelatase deficiency which is the last step in heme synthesis ❼ hence water solubility will be the least ❼ porphyrins cannot be detected in the urine.

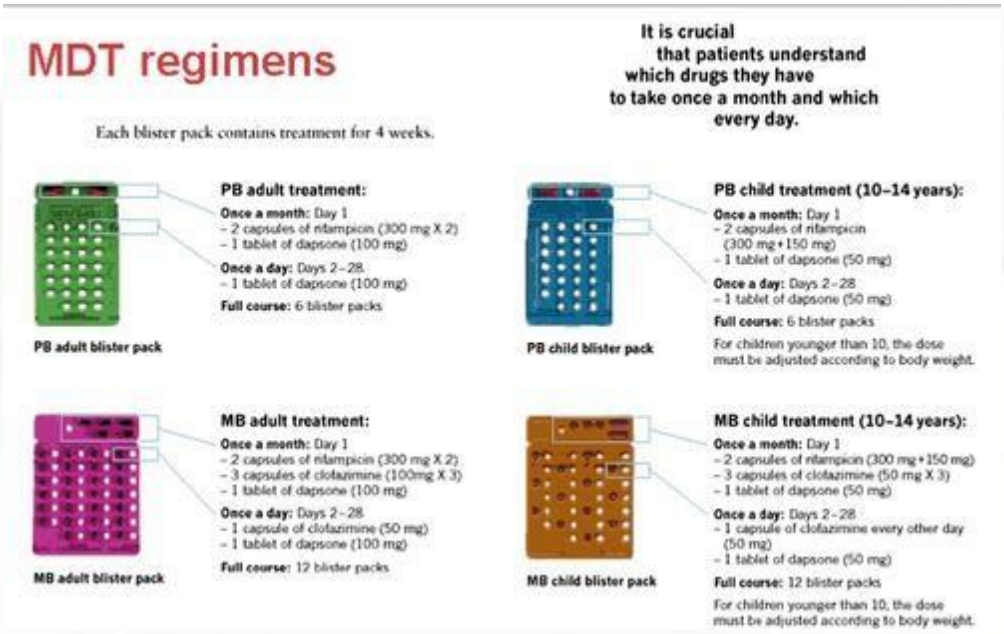
20) A 24 year old male presents with a solitary hypo-pigmented, hypo-anesthetic patch over forearm, nerves were not enlarged, slit skin smear showed 2+ which treatment is appropriate?



A) Red blister pack
B) Green blister pack
C) Yellow blister pack
D) Blue blister pack

✅ Option A- Red blister pack

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- In the above question the patient is a 24 yr old hence the adult pack is chosen.
- On SSS examination is his bacteriological index is 2+, hence multibacillary blister pack is given. On SSS examination even if a single bacilli is found it is considered as multibacillary leprosy.
- Green pack: PB adult
- Blue pack: PB child
- Red pack: MB child
- Yellow pack: MB child

21) A young male presented with itchy skin lesion over the trunk with central clearing, the causative agent for the lesion is?



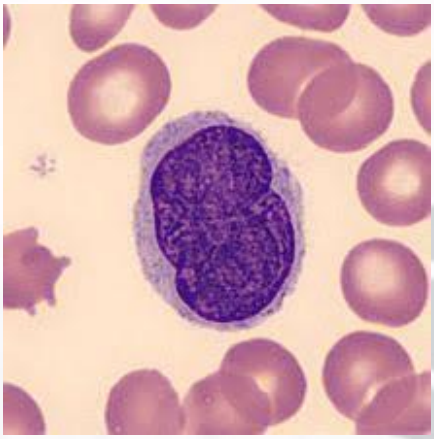
A) Trichophyton rubrum
B) Trichophyton concentricum
C) Trichophyton mentagrophytes
D) Trichophyton violaceum

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✅ Option B- Trichophyton concentricum

- The image is of Tinea imbricate or Tokelau which is a variant of tinea corporis caused by T. concentricum.
- Characteristic pattern of concentric plaques of erythema and scaling.

22) A 61-year-old African American woman with a past medical history of C6-7 spinal stenosis complained of a pruritic rash that began on her face one year prior. The rash appeared raised and erythematous and contained plaques and patches. Despite treatment with topical steroids and emollients, these lesions evolved into cutaneous tumors and spread to encompass >50% of her body surface. Peripheral blood examination showed the following cells. What is your diagnosis?



A) Hodgkins lymphoma
B) Sezary syndrome
C) Leukemia cutis
D) None of the above

✅ Option B- Sezary syndrome

- The history given is suggestive of malignant etiology as the patient is of elderly age group, persistant lesion which is not responding to treatment.
- The peripheral blood examination shows a picture of an atypical lymphocyte with cerebriform nucleus. The cytoplasm is sparse and present as a peripheral rim around the nucleus. This is characteristic of ***SEZARY CELL/ MONSTER CELL.***
- Increased number of sezary cells are seen in mycosis fungoides and sezary syndrome.
- Reed Sternberg cells are seen in Hodgkin's lymphoma.

23) young male presents with multiple, minute, intensely pruritic wheals to papules 1-3mm in diameter with erythematous flare, they subside within 30-90 minutes. What is the most likely diagnosis?

A) Urticaria pigmentosa
B) Urticarial vasculitis

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C) Angioedema

D) Cholinergic urticaria

1. ✅ Option D – Cholinergic urticaria
   - Cholinergic urticaria is a type of urticarial that is precipitated by increase in core body temperature. Triggers include exercise, sweating, hot and spicy foods.

     The wheals are small, punctate 1-4mm in size.
   - Urticarial vasculitis presents as serpentine wheals with burning sensation and are often tender. They last for more than 24 hours and leave behind residual hyperpigmentation.
   - Urticarial pigmentosa is a type of cutaneous mastocytosis. Seen usually in children.
   - Angioedema is characterised by painless, non-pruritic, non-pitting swelling of submucosal, dermal or subcutaneous tissue.

24) A 26 year old male presents with patchy loss of hair over the scalp, on trichoscopy the following finding was present. What is your likely diagnosis??



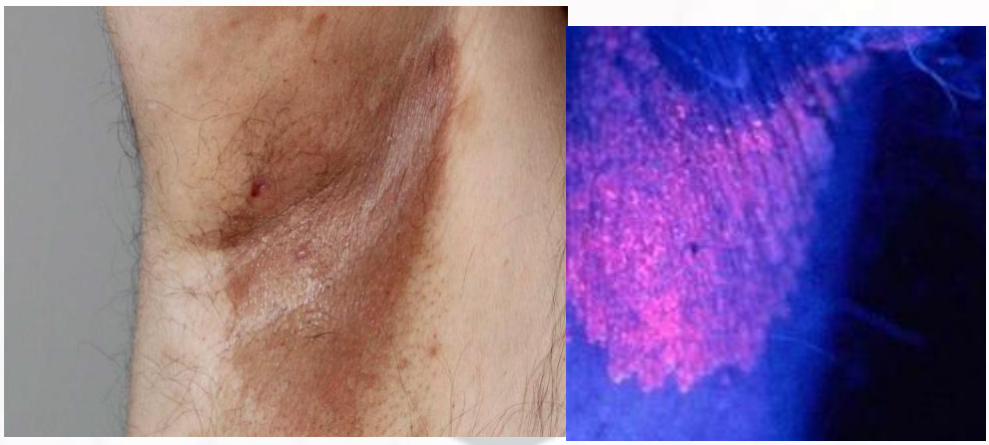
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A) Tinea capitis
B) Alopecia areata
C) Lichen planopilaris
D) Frontal fibrosing alopecia

✅ Option B- Alopecia areata

- The image shows characteristic exclamatory hair on dermascopy.
- These are seen around the patches of alopecia. They are short hair that are tapered proximally.

25) A young male presents with a hyper-pigmented patch over his axilla. Clinical picture and woods lamp examination revealed the following findings. What is the causative agent for this condition?



A) Corynebacterium minutissimum
B) Ptyriasis versicolor
C) Corynebacterium ulcerans
D) Corynebacterium xerosis

2. ✅ Option A- Cornybacterium minutissimum

- The history, clinical image and wood's lamp examination is suggestive of Erythrasma.
- Erythrasma is caused by Cornybacterium minutissimum.
- It favours flexural sites with moist environment.
- Coral red fluorescence on Wood's lamp examination is diagnostic. It occurs due to coproporphyrin produced by the bacteria.

26) A 30 year old male presented with multiple fluid filled blisters all over the body including the scalp with mucous membrane involvement, which ruptures spontaneously to form raw painful erosions. No history of any drug intake, photosensitivity or application of any irritant topical medication. No h/o exacerbation on alcohol consumption. No h/o weight loss. The patient is not willing for skin

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biopsy to confirm the diagnosis, which other specimen will help you in the diagnosis of the above condition?

A) Nail

B) Saliva

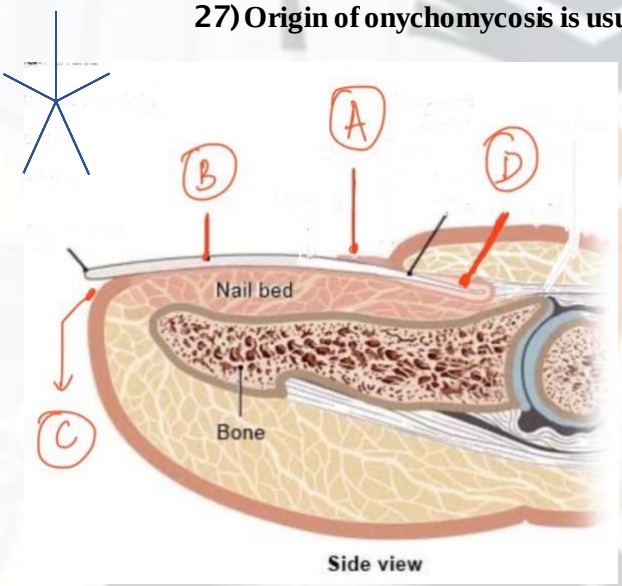
C) Hair

D) Urine

☑ Option C – Hair

- The question describes classically a patient of pemphigus vulgaris. A negative history of -drug intake rules out bullous drug eruptions
  -photosensitivity rules out bullous SLE
  - topical application rules out irritant contact dermatitis
  - exacerbation on alcohol rules out porphyria cutanea tarda.
  Since it is present all over the body and not confined to trauma prone areas epidermolysis bullosa acquisita can be ruled out.
- Direct immunofluorescence of the outer root sheath of telogen hair can be used for the diagnosis and follow up for patients with pemphigus.
- Desmogleins (1, 3 and 4) are expressed in the hair.

27) Origin of onychomycosis is usually from-



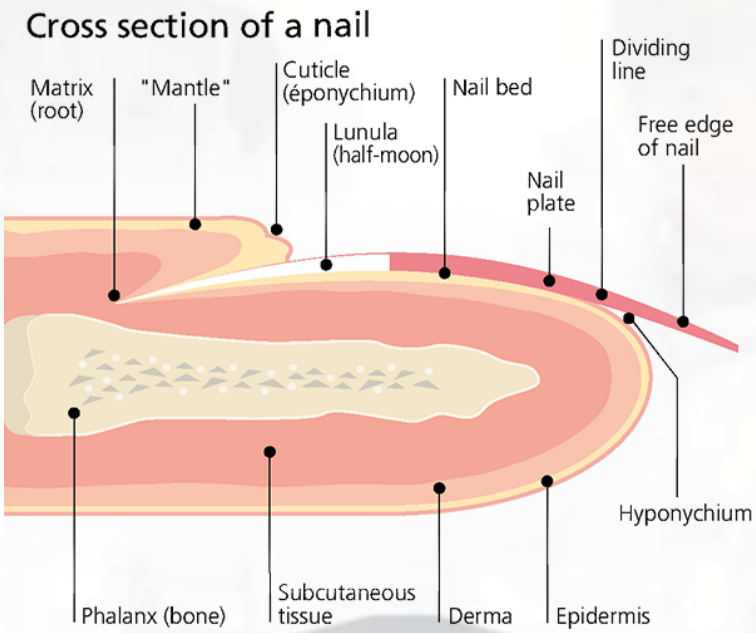
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A) A
B) B
C) C
D) D

☑ Option C

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- Onychomycosis usually originates from the hyponychium which is labelled as C.



- A : eponychium
- B: nail plate
- C: hyponychium
- D: nail matrix

★ **28) Matrix metallo proteinases (MMP) are calcium dependent zinc containing enzymes. Homeostasis between the matrix metalloproteinases (MMP) and tissue inhibitor of matrix metallo proteinases is important for wound repair and healing. If there is over activity of tissue inhibitor of matrix metallo proteinases what would be the likely outcome during wound healing?**

A) Wound dehiscence
B) Hypertrophic scars
C) Delayed wound healing
D) Scanty granulation tissue.

✓ Option B –Hypertrophic scar

- MMP are a group of enzymes that degrade extra cellular matrix (like collagen) in the dermis.
- These enzymes are regulated by tissue inhibitor of MMP.
- Therefore, if there is over activity of tissue inhibitor of M M P ❼ decrease activity of MMP ❼ accumulation of extracellular matrix (collagen) ❼ hypertrophic scar.

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29) All of the following are manifestations of late congenital syphilis except

A) Eight nerve deafness
B) Sabre tibia
C) Clutton joint
D) Saw tooth metaphyses on x ray

✅ Option D – Saw tooth metaphysis on X ray

- Saw tooth metaphysis occurs as a result of osteochondritis. In the medial aspect of the proximal tibial metaphysis, this defect has been called the WIMBERGER'S OR CATBITE sign.

Early Disease	*Late Disease*
• Hepatosplenomegaly • Jaundice • Thrombocytopenia • Mucocutaneous rash, often involving the palms and soles • Condylomatous lesions • "Snuffles" (rhinitis) • Radiographic changes—periostitis and metaphyseal dystrophy • Osteochondritis and pseudoparalysis • Lymphadenopathy • Failure to thrive • Chorioretinitis • Renal abnormalities	• Prominence of forehead ("Olympian brow") • Scaphoid scapula • Tibial bowing ("saber shins") • Peg-shaped, notched teeth ("Hutchinson teeth") • Mulberry molars • Saddle nose • Rhagades (linear scars) • Painless arthritis of knees ("Clutton joints") • Interstitial keratitis (which may lead to blindness) • Eighth cranial nerve deafness • Juvenile paresis • Juvenile tabes

30) A three year old male child presented with large, dark scales, predominantly involving the flexures of the body since neonatal period. There was H/O difficulty in initiation of labour in the mother. Slit-lamp examination of the child revealed corneal opacities. The most probable diagnosis would be

A) Ichthyosis vulgaris
B) Congenital icthyosiform erythroderma
C) X-linked recessive ichthyosis
D) Harlequin ichthyosis.

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✅ Option C- X linked ichthyosis

- X linked ichthyosis is caused by mutation in steroid sulfatase (STS) enzyme.
- This enzyme is also required for hydrolysis of DHEAS ➡ necessary for estrogen synthesis.
- Therefore if STS is deficient in fetal placenta there will be low estrogen levels in the urine and amniotic fluid ➡ insufficient dilation of cervix ➡ labour fails to intiate spontaneously or progress.
- X linked ichthyosis is also unique because it has extra cutaneous manifestations like corneal opacities and cryptorchidism.
- Ichthyosis vulgaris has <u>finer scales and flexures are spared</u>. There will be no extra cutaneous involvement.
- Congenital ichthyosiform erythroderma- the scales will be of powdery consistency and erythroderma will be prominent.
- Harlequin ichthyosis – it is a severe form. Baby will have armor like thick scales which encases the baby. Associated with ectropion, eclabium and autoamputation.

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